



Sales Performance Analysis Report

Project Overview

The purpose of this project is to analyse sales performance across regions, product categories, and time to identify best- and worst-performing products and uncover seasonal patterns. The analysis aims to support data-driven business decisions related to sales strategy, inventory planning, and regional growth.

Objectives

- Analyse overall sales performance
 - Identify top and low-performing categories and sub-categories
 - Compare sales across regions
 - Analyse monthly and yearly sales trends
 - Detect seasonality in sales
 - Provide actionable business recommendations
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Dataset Description

The dataset contains **9,800 rows and 18 columns**, including:

- Order and shipping details
- Customer information
- Product category and sub-category
- Regional data
- Sales values

Key Columns Used

- **Order Date** (time-based analysis)
 - **Region** (regional performance)
 - **Category & Sub Category** (product analysis)
 - **Sales** (revenue metric)
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Data Cleaning & Preparation

The following preprocessing steps were performed:

- Converted Order Date and Ship Date to datetime format
- Checked and validated data types
- Identified missing values
 - Postal Code had 11 missing values
 - Sales had no missing or invalid values
- Verified that no sales values were zero or negative

The cleaned dataset was then used for analysis and visualization.

Key Performance Indicators (KPIs)

KPI	Value
Total Sales	\$2,261,536.78
Total Unique Orders	4,922
Average Order Value (AOV)	\$459.48

These KPIs help evaluate overall revenue performance and customer purchasing behaviour.

Sales Analysis & Insights

Sales by Category

- **Technology** generated the highest revenue
- Followed by **Furniture** and **Office Supplies**

This indicates strong customer demand for technology-related products.

Top Sub-Categories by Sales

Top-performing sub-categories:

1. Phones
2. Chairs
3. Storage

4. Tables
5. Binders

Phones and Chairs contributed the most revenue, suggesting they are key drivers of sales.

Regional Sales Performance

Sales ranking by region:

1. **West** – Highest revenue
2. East
3. Central
4. South – Lowest revenue

The West region is the strongest market, while the South shows potential for improvement.

Monthly Sales Trend

- Highest sales month: **February 2017**
- Lowest sales month: **March 2015**

Overall, sales showed an upward trend over the years with noticeable monthly fluctuations.

Seasonality Analysis

Average sales by month revealed:

- **Best month (average): August**
- **Worst month (average): May**

This confirms seasonal patterns in customer purchasing behaviour.

Visualizations Created

- Monthly Sales Trend Line Chart
- Total Sales by Category (Bar Chart)
- Top Sub-Categories by Total Sales
- Regional Sales Comparison

These visuals helped clearly communicate trends and performance differences.

Business Recommendations (Tactical Improvements)

1. Increase inventory and promotions for high-performing sub-categories like Phones and Chairs
 2. Launch targeted marketing campaigns in underperforming regions such as the South
 3. Leverage peak months (August) for major sales campaigns
 4. Introduce bundled offers in Technology and Furniture categories
 5. Analyse customer behaviour further to improve repeat purchases and order value
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Limitations

- Profit and cost data were not available
 - Customer conversion metrics could not be calculated
 - Analysis focused on historical sales data only
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Conclusion

This project successfully analysed sales performance across multiple dimensions. The insights highlight strong product categories, regional opportunities, and clear seasonality trends. The findings can help businesses optimize sales strategies, improve regional focus, and plan seasonal promotions more effectively.

Source code

Here is the link of [Source Code](#)